

PERSONAL FINANCE TRACKER

(A SERVERLESS AWS SOLUTION FOR PERSONAL FINANCIAL MANAGEMENT)

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AGENDA

- Problem Statement
- Solution Overview
- Key Features
- AWS Services Used
- Architecture
- Future Work
- Conclusion

PROBLEM STATEMENT

- Manual tracking with spreadsheets or notebooks is tedious and error-prone
- People lose visibility into where their money goes each month
- No real-time view of income vs. expense balance
- Existing tools often require installing an app or maintaining a server

SOLUTION OVERVIEW

- A web-based Personal Finance Tracker, fully hosted on AWS with no physical server
- Users add income or expense transactions via a simple web form (amount, type, category, date, description)
- The FinanceTrackeFunction Lambda validates and stores each transaction in Amazon DynamoDB
- A GetTransactions Lambda retrieves records to power the Transaction History table and dashboard
- UpdateTransaction and DeleteTransaction Lambdas let users edit or remove entries
- The dashboard shows Total Income, Total Expense, and Balance, plus trend and category charts
- The entire frontend is hosted as a static website on Amazon S3

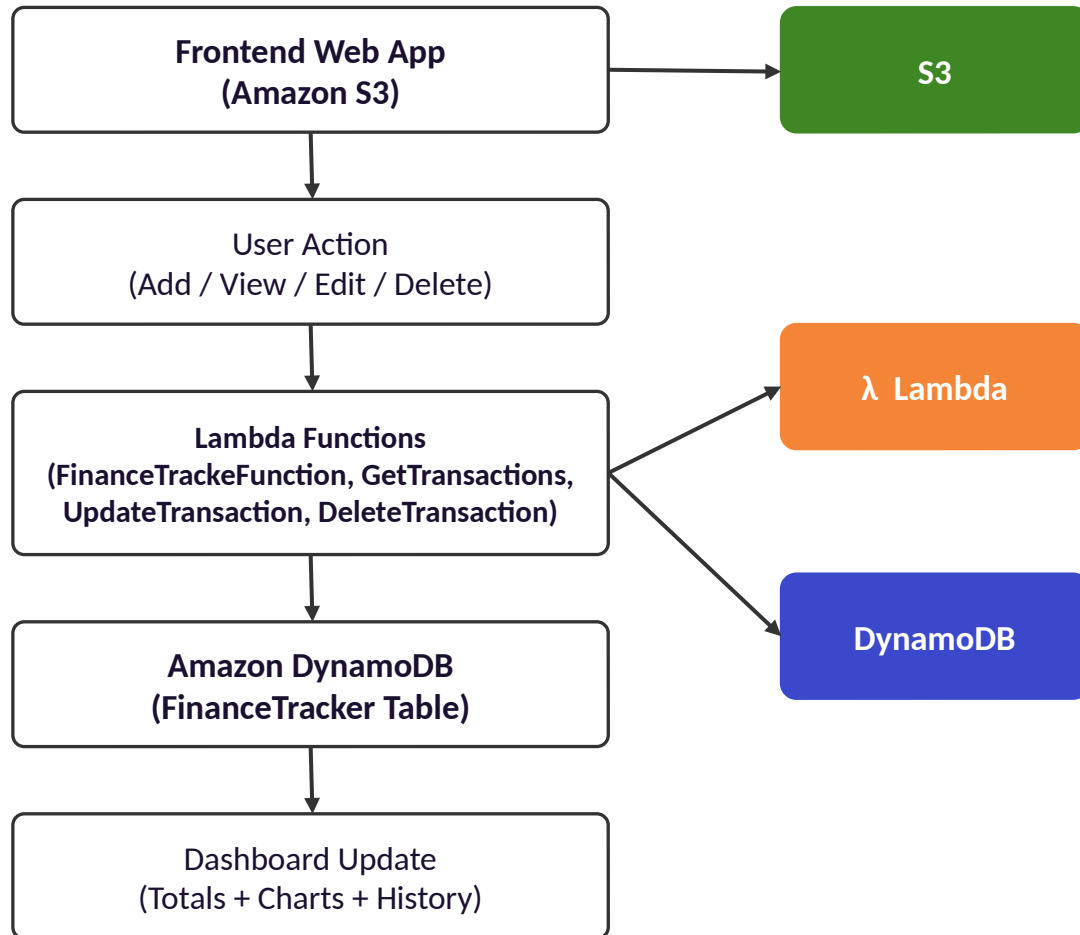
KEY FEATURES

1. Add income / expense transactions through a simple web form
2. Edit and delete existing transactions
3. Automatic categorization (Food, Shopping, Salary, etc.)
4. Real-time dashboard — Total Income, Total Expense, Balance
5. Monthly Expense Trend line chart
6. Expense Analysis pie chart by category
7. Fully serverless backend using AWS Lambda + DynamoDB
8. Frontend hosted on an Amazon S3 static website

AWS SERVICES USED

AWS Service	Purpose
Amazon S3	Hosts the static frontend dashboard as a public website
AWS Lambda	Runs FinanceTrackeFunction, GetTransactions, UpdateTransaction, DeleteTransaction
Amazon DynamoDB	Stores transaction records (user ID, TransactionID, amount, type, category, date, description)
AWS IAM	Grants least-privilege access between services
Amazon CloudWatch	Monitors Lambda execution logs and performance metrics

ARCHITECTURE



FUTURE WORK

- Mobile app integration for on-the-go tracking
- Budget-limit alerts via Amazon SNS
- Multi-user authentication using Amazon Cognito
- Predictive spending analytics
- Exportable monthly reports (PDF / Excel)

CONCLUSION

- A secure, automated, and scalable personal finance solution
- Eliminates manual tracking errors and spreadsheet clutter
- Fully serverless architecture reduces cost and operational overhead
- Gives real-time visibility into income, expenses, and balance

THANK YOU